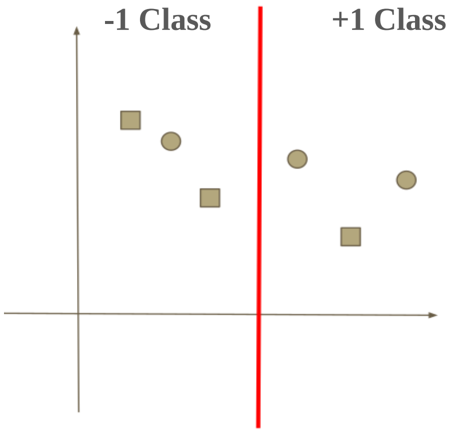
An Example of AdaBoost Algorithm

Here,
y = Actual Label
$h^1(x)$ = Prediction of first weak classifier

Update the weights of each data points.

Features	**y**	**$h^1(x)$**	**$h^2(x)$**
f1	+1	+1	-1
f2	+1	-1	+1
f3	+1	+1	+1
f4	-1	-1	+1
f5	-1	+1	-1
f6	-1	-1	-1



An Example of AdaBoost Algorithm

Solution:

Features	y	$h^1(x)$	Remarks
f1	+1	+1	Correct
f2	+1	-1	Wrong
f3	+1	+1	Correct
f4	-1	-1	Correct
f5	-1	+1	Wrong
f6	-1	-1	Correct

Number of data points = 6

Therefore, Initial Weights, $W^t = \frac{1}{6}$; for each data points

Error at step "t", $e^t = 2/6 = \frac{1}{3}$

Updating weights, $$w_i^{t+1} = \frac{w_i^t}{2} \times \begin{cases} \frac{1}{1-e^t} & \text{correct} \\ \frac{1}{e^t} & \text{wrong} \end{cases}$$

For correct-predicted datapoints, $W^{t+1} = (\frac{1}{6})/2 * 1/(1- \frac{1}{3})$

$= \frac{1}{8}$

For wrong-predicted datapoints, $W^{t+1} = (\frac{1}{6})/2 * 1/(\frac{1}{3})$

$= \frac{1}{4}$

An Example of AdaBoost Algorithm

After first iteration:

Features	y	$h^1(x)$	Updated Weights
f1	+1	+1	1/8
f2	+1	-1	1/4
f3	+1	+1	1/8
f4	-1	-1	1/8
f5	-1	+1	1/4
f6	-1	-1	1/8

An Example of AdaBoost Algorithm

Solution: For Second Iteration

Number of data points = 6
Therefore, For correct-predicted datapoints, $W^{t+1} = 1/8$;
For wrong-predicted datapoints, Wt+1 = (1/6)/2 * 1/(1/3) = 1/4

Error after 1st iteration, e^t = f1+f4=(1/8)+(1/8) = 1/4

Updating weights,
$$w_i^{t+2} = \frac{w_i^{t+1}}{2} \times \begin{cases} \frac{1}{1-e^t} & \text{correct} \\ \frac{1}{e^t} & \text{wrong} \end{cases}$$

For correct-predicted datapoints(f2), W^{t+2} = (1/4)/2 * 1/(1- 1/4)
= 1/6
For correct-predicted datapoints(f3), W^{t+2} = (1/8)/2 * 1/(1- 1/4)
= 1/12

Features	y	$h^2(x)$	Remarks	Updated Weights
f1	+1	-1	Wrong	1/8
f2	+1	+1	Correct	1/4
f3	+1	+1	Correct	1/8
f4	-1	+1	Wrong	1/8
f5	-1	-1	Correct	1/4
f6	-1	-1	Correct	1/8

An Example of AdaBoost Algorithm

Solution: For Second Iteration

For correct-predicted datapoints(f5), $W^{t+2} = (1/4)/2 * 1/(1- ¼)$
$= 1/6$

For correct-predicted datapoints(f6), $W^{t+2} = (1/8)/2 * 1/(1- ¼)$
$= 1/12$

For wrong-predicted datapoints(f1), $W^{t+2} = (1/8)/2 * 1/(¼)$
$= 1/4$

For wrong-predicted datapoints(f4), $W^{t+2} = (1/8)/2 * 1/(¼)$
$= 1/4$

Features	y	$h^2(x)$	Remarks	Updated Weights
f1	+1	-1	Wrong	1/8
f2	+1	+1	Correct	1/4
f3	+1	+1	Correct	1/8
f4	-1	+1	Wrong	1/8
f5	-1	-1	Correct	1/4
f6	-1	-1	Correct	1/8

An Example of AdaBoost Algorithm

After second iteration:

Features	y	$h^2(x)$	Updated Weights
f1	+1	-1	1/4
f2	+1	+1	1/6
f3	+1	+1	1/12
f4	-1	+1	1/4
f5	-1	-1	1/6
f6	-1	-1	1/12